LOCALITY-AWARE ACTIVATION INTERVENTION FOR MULTI-HOP FACTUAL ADAPTATION IN LARGE LANGUAGE MODELS

Anonymous Authors

Affiliation withheld for the current draft

ABSTRACT

Large language models can absorb counterfactual facts without changing parameters, yet multi-hop updates may damage unrelated outputs. We study Value-CQAG, a content-routed activation intervention that injects an explicit-fact contrast only when a question overlaps the edited context. On a frozen 100-case, 300-question MQuAKE-CF confirmation pool, question-level generation rises from 2.00% to 38.67% on Qwen2.5-7B and from 4.00% to 51.33% on Llama-3.1-8B. A separate 10,000-pair audit gives 98.49% exact-output locality. We compare explicit facts, retrieval-plus-in-context learning, and parameter editing with MEMIT. Retrieval reaches 95.0% question-level generation on Qwen but has lower locality; MEMIT reaches 16.0% and 14.33% on Qwen and Llama. Failure audits distinguish router misses, old-answer residuals, and multi-hop entity errors. The study presents activation intervention as a selective, auditable adaptation path rather than an accuracy or knowledge-editing state-of-the-art claim.

Index Terms— large language models, factual adaptation, activation intervention, locality, multi-hop reasoning, generative AI

1. INTRODUCTION

Large language models (LLMs) store factual associations in distributed internal representations and can answer multi-hop questions by composing those associations. Updating one fact, however, can expose a difficult systems trade-off: the intervention must affect questions that depend on the edited fact while leaving unrelated generations unchanged. This is especially important for applications that require rapid, reversible adaptation without modifying model weights.

Existing knowledge-editing methods modify parameters directly, for example by locating and rewriting factual associations in feed-forward layers [1–3], or by storing an external edit memory [4]. Multi-hop evaluation is important because single-hop success does not imply that the updated fact is used in a composed chain [5]. Retrieval and in-context methods avoid parameter changes but can introduce long prompts and broad interference [6, 7]. These comparisons are often made using a single success number, which obscures whether a failure comes from missing the relevant context, retaining the old answer, or choosing an incorrect entity during multi-hop composition.

We present a controlled study of Value-CQAG, a value-level activation intervention with a content-token router. The router computes overlap between the current question and the edited context; only questions above a fixed threshold receive the intervention. The method is deliberately conservative: the model remains frozen, the intervention is prompt-only, and locality is measured on a separate question set. Figure 1 summarizes the evaluation path. This separation of routing, intervention, and generation makes the method amenable to signal-path analysis: false routes can be measured independently from downstream multi-hop reasoning failures.

Our contributions are:

- A locality-aware activation intervention for multi-hop factual adaptation that separates routing from generation and does not update model weights.
- A controlled cross-model evaluation on Qwen2.5-7B and Llama-3.1-8B with the same frozen confirmation pool, explicit-fact and retrieval controls, and MEMIT baselines.
- A 10,000-pair end-to-end locality audit and a failure decomposition that distinguishes router misses, old-answer residuals, and entity/reasoning errors.

2. METHOD

2.1. Problem setup

Let $\mathcal{E} = \{(s_i, r_i, o_i^{\text{new}})\}_{i=1}^{K}$ be a set of edited subject–relation–object facts and let q be a question whose answer may require several facts. The base model f_θ is frozen. We seek a conditional intervention $\Delta(q, \mathcal{E})$ such that

$$\hat{y} = f_\theta(q; \Delta(q, \mathcal{E})) \tag{1}$$

is consistent with the edited facts, while $f_\theta(q; 0)$ is retained for unrelated questions. This differs from parameter editing, which produces a single edited model $f_{\theta+\Delta\theta}$ for all subsequent inputs. The desired behavior is therefore conditional, not a global change of the model's memory.

2.2. Activation target and gated operator

Let $H_l^0(q) \in \mathbb{R}^{T \times d_l}$ denote the layer-l hidden states when the question is encoded alone, and let $H_l^+(q, \mathcal{E})$ denote the states when the same question is encoded with a compact prompt containing the edited facts. We form a teacher-free, answer-independent activation target

$$D_l(q, \mathcal{E}) = H_l^+(q, \mathcal{E}) - H_l^0(q). \tag{2}$$

This difference uses only the question and the edited triples; no target answer or generated completion enters D_l. Let P_w select the final w prompt positions and let $g(q, \mathcal{E}) \in \{0, 1\}$ be the router decision. The injected states at the selected layers $\mathcal{L}$ are

$$\widetilde{H}_l = H_l^0 + g(q, \mathcal{E})\, \alpha P_w D_l, \qquad l \in \mathcal{L}. \tag{3}$$

The confirmation configuration uses $\mathcal{L} = \{20, 22, 24, 26\}$, $w = 16$, and $\alpha = 5$; these values are selected on development cases and frozen before the 100-case confirmation. Equation 3 is applied during the prompt encoding pass only. The model weights and all later autoregressive decoding states remain unchanged.

The construction is related to recent activation steering and representation intervention methods [8, 9], but differs in two ways: the direction is obtained from an explicit-fact contrast rather than a learned human-preference direction, and a content router gates the direction on a query-by-query basis.

2.3. Content-token routing

We represent the edit context and the question as sets of normalized content tokens, removing stop words and punctuation. The router score is the Jaccard overlap

$$\rho(q, \mathcal{E}) = \frac{|T(q) \cap T(\mathcal{E})|}{|T(q) \cup T(\mathcal{E})|}. \tag{4}$$

The gate is

$$g(q, \mathcal{E}) = \mathbb{I}[\rho(q, \mathcal{E}) \geq \tau], \qquad \tau = 0.35. \tag{5}$$

The threshold is fixed on development data and is not tuned on the confirmation pool. A routed miss is counted separately from a generation error after a route. This explicit abstention mechanism is the reason we report router coverage and generation conditional on routing rather than one conflated accuracy number.

2.4. First-order effect and locality identity

Let $z(q)$ be the pre-softmax logits and define the new-answer margin $m(q) = z_{y_{\text{new}}}(q) - z_{y_{\text{old}}}(q)$. A first-order expansion around the unmodified hidden states gives

$$\Delta m \approx \alpha g \sum_{l \in \mathcal{L}} \langle \nabla_{H_l} m, P_w D_l \rangle + O\left(\alpha^2 \sum_l \|P_w D_l\|_F^2 \right). \tag{6}$$

Thus the intervention improves the new-answer margin when the explicit-fact direction is aligned with the local margin gradient, while a random direction has expectation near zero under an isotropy assumption. This is a diagnostic model, not a claim of a global guarantee: the measured failure categories test when the alignment assumption breaks during multi-hop composition.

The router also gives an exact locality decomposition. For an unrelated query u, let $\phi = \Pr[g(u, \mathcal{E}) = 1]$ be the false-positive route rate and let $\eta = \Pr[\text{output changes} \mid g = 1]$. Because rejected queries bypass Equation 3,

$$L_{\text{exact}} = 1 - \phi\eta. \tag{7}$$

On the 10,000-pair audit, $\phi = 152/10{,}000 = 1.52\%$ and $\eta = 0.9934$, yielding $L_{\text{exact}} = 98.49\%$. Equation 7 explains why a high router rejection rate is necessary but not sufficient: a false route is highly likely to alter the output.

2.5. Two-stage prompt-pass implementation

Figure 1 makes the conditional computation explicit. In Stage 1, two frozen prompt passes encode the same question: a question-only pass produces $H_l^0(q)$, while an edited-fact-conditioned pass produces $H_l^+(q, \mathcal{E})$. The router independently tests whether the question is relevant to the edit. In Stage 2, an inactive route forwards $H_l^0(q)$ unchanged to the shared frozen decoder. An active route first computes $D_l(q, \mathcal{E}) = H_l^+(q, \mathcal{E}) - H_l^0(q)$ and then supplies $\widetilde{H}_l = H_l^0 + g(q, \mathcal{E})\alpha P_w D_l$ at the selected prompt positions before decoding. The pictorial input symbols are schematic; all reported experiments use text questions and text-form edited facts.

The intervention is applied only during the prompt encoding pass and is not repeatedly injected into later autoregressive steps; decoder weights remain frozen in both routes. This choice is empirical: continuing the intervention for the first two generated tokens decreases question-level generation by 4.00 percentage points on Qwen and 7.00 points on Llama. Prompt-only intervention therefore provides the primary method reported here.

3. RELATED WORK

Knowledge editing has been studied through localized parameter updates such as ROME and MEMIT [2, 3], and through external edit memories such as SERAC [4]. The MQuAKE benchmark specifically exposes failures that only appear when several edited facts must be composed [5]. Our intervention leaves the parameters and external memory unchanged and instead controls the activation signal presented to a frozen decoder.

Activation steering methods add directions at inference time, including unsupervised activation addition [8] and inference-time interventions for truthful behavior [9]. These methods motivate treating hidden states as a controllable signal, but they generally assume a direction is applied whenever a prompt is evaluated. Our contribution is the combination of an explicit-fact contrast, a lexical gate, and a separate end-to-end locality audit. Retrieval-augmented generation remains a strong accuracy control [6]; we therefore report it rather than presenting activation intervention as an accuracy replacement.

4. EXPERIMENTAL PROTOCOL

4.1. Models and data

We evaluate Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct. All principal comparisons use the same frozen 100-case MQuAKE-CF confirmation pool, containing 300 generated questions; MQuAKE was designed to test multi-hop knowledge editing rather than isolated single facts [5]. The pool is kept separate from a reserved 60-case set that is not used in this study. For each edited case, five deterministic unrelated questions are used for the parameter-editing locality evaluation.

4.2. Baselines and metrics

We compare four adaptation modes: (i) the unmodified base model, (ii) explicit facts placed in the prompt, (iii) a trans-

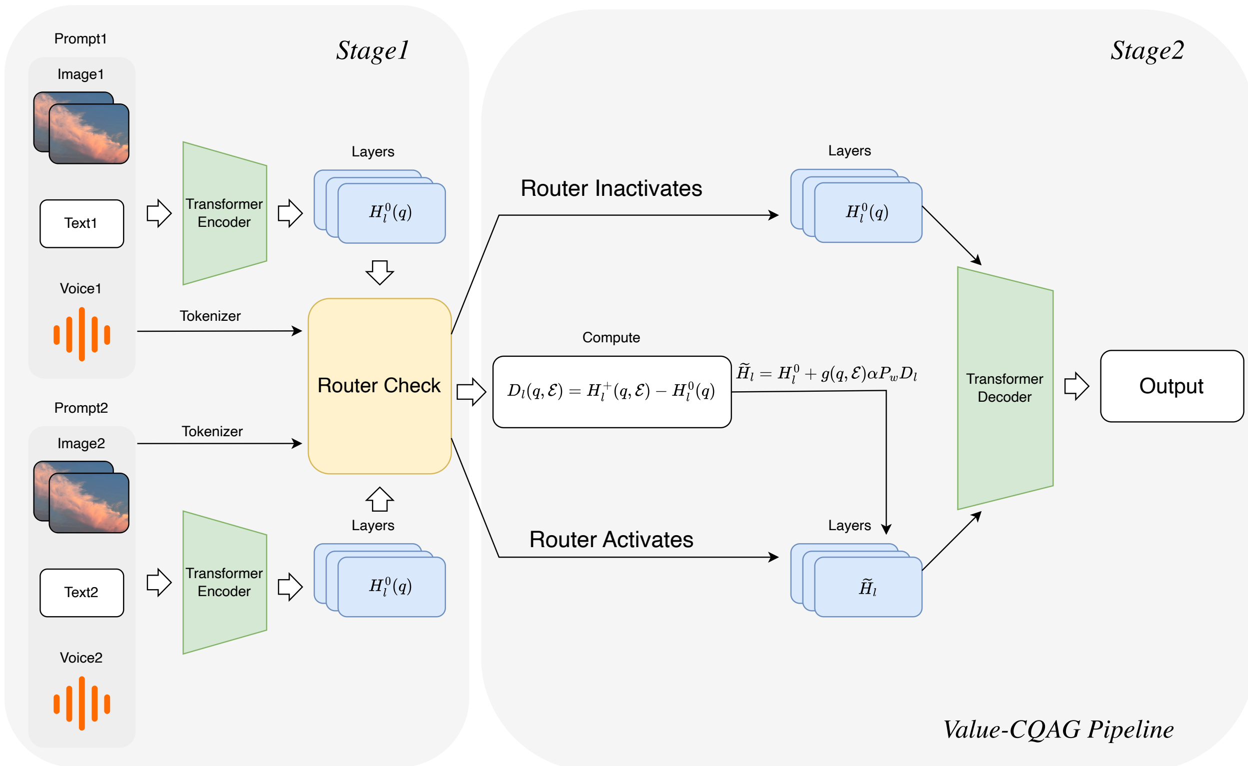

Fig. 1. Two-stage Value-CQAG pipeline. Stage 1 encodes question-only and edited-fact-conditioned prompts to obtain $H_l^0(q)$ and $H_l^+(q,\mathcal{E})$, while the router tests edit relevance. In Stage 2, rejected queries retain $H_l^0(q)$; accepted queries receive the contrast $D_l(q,\mathcal{E})$ through $\widetilde{H}_l$ before frozen decoding.

parent retrieval-plus-ICL control inspired by IKE [7], and (iv) EasyEdit MEMIT [3]. The retrieval corpus contains 500 non-overlapping development cases and uses MiniLM semantic retrieval with eight demonstrations. This is an IKE-style control, not a claim of an unmodified official IKE reproduction.

We report question-level generation accuracy, case-level generation accuracy (all questions in a case correct), new-answer preference, and exact-output locality. The routed locality audit contains 100 edited cases crossed with 100 unrelated questions, yielding 10,000 pairs. Exact locality requires an unchanged output string; semantic locality uses the task's semantic equivalence check. For MEMIT, locality is measured on 500 fixed unrelated outputs and is not directly pooled with the 10,000-pair router audit.

4.3. Reproducibility

For MEMIT, second-moment statistics use the local 'wikimedia/wikipedia' 20231101.en mirror, 100,000 samples, float32, and layers 4–8 of 'mlp.down_proj'. Due to the 24-GB RTX 3090 memory limit, models are loaded in fp16; the covariance linear system is solved in CPU double precision and only the resulting update is moved to the GPU. The scripts, hparams, frozen IDs, and raw JSON artifacts accompany this paper draft.

Table 1. Question-level and case-level generation accuracy on the same frozen confirmation pool. Retrieval+ICL† was run on Qwen only, using eight demonstrations from 500 development cases.

Model	Method	Question	Case
Qwen2.5-7B	Base	2.00%	0.0%
	Value-CQAG	38.67%	18.0%
	Explicit facts	95.00%	90.0%
	Retrieval+ICL†	95.00%	93.0%
Llama-3.1-8B	Base	4.00%	0.0%
	Value-CQAG	51.33%	32.0%
	Explicit facts	93.00%	87.0%

5. RESULTS

5.1. Cross-model comparison

Table 1 shows the primary confirmation results. Value-CQAG improves over the base model on both architectures, but explicit facts and retrieval remain stronger on answer accuracy. This separation is useful: the contribution is controlled adaptation and locality, not a claim that a short intervention prompt outperforms a large retrieval context.

Relative to the base model, the paired question-level gain is +36.67 percentage points for Qwen (95% CI [29.33, 44.33]) and +47.33 points for Llama (95% CI [39.33, 55.33]).

Table 2. EasyEdit MEMIT on the frozen 100-case pool. Locality uses 500 unrelated outputs.

Model	Pref.	Q-gen.	Case	Exact loc.	Edit (s)
Qwen2.5-7B	16.0%	16.0%	8.0%	85.2%	52.41
Llama-3.1-8B	18.0%	14.33%	5.0%	73.4%	29.64

Table 3. Completed auxiliary evaluations. "Locality" is mismatched-direction locality for Value-CQAG and exact-output locality for ROME. Results are not pooled across protocols.

Setting	n	Q-gen.	Case	Locality	Seeds
Qwen CF, Value-CQAG	30×3	27.78±2.22	12.22±1.92	80.00±14.53	3
Qwen T, Value-CQAG	16×3	21.53±2.41	0.00	64.58±7.22	3
Qwen2.5-3B CF	30	3.33	0.00	–	1
Llama CF, question-specific	80	42.92	28.75	–	1
Qwen CF, ROME	80	19.17	13.75	80.75	1

Retrieval-plus-ICL reaches 95.0% question-level and 93.0% case-level accuracy on Qwen, but its exact and semantic locality are only 25.0% and 61.4%, respectively.

5.2. Locality and parameter editing

The end-to-end routed audit on Qwen gives a 1.52% false-positive rate, 98.49% exact-output locality, and 99.62% semantic locality. Conditional on a false route, exact locality falls to 0.66%, showing that the router is the main safety boundary. The audit records 37 newly corrupted originally-correct outputs and only one newly fixed originally-incorrect output.

Table 2 compares the two MEMIT runs. The method has lower generation accuracy than Value-CQAG and substantially lower locality on Llama. These results indicate that a weight edit can provide a persistent change, but persistence is not equivalent to selective behavior on unrelated questions.

5.3. Failure decomposition

Among 300 Qwen questions, 116 are correct, 133 are other-entity or reasoning errors, 31 retain the old answer, 19 are router misses, and one is a refusal. Llama yields 154 correct, 96 other-entity or reasoning errors, 30 old-answer residuals, 19 router misses, and one refusal. The identical router-miss count across models separates coverage from the downstream generation bottleneck: after a successful route, multi-hop entity selection remains the dominant failure mode.

5.4. Additional completed experiments

Table 3 reports completed auxiliary results. Their case pools and filtering rules differ from the confirmation protocol, so they are not pooled.

The explicit-fact control reaches 95.0% / 90.0% question/case accuracy on Qwen and 93.0% / 87.0% on Llama. On the earlier 30-case split it reaches 78.89% on Qwen2.5-7B and 74.44% on Qwen2.5-3B, establishing that the data are solvable.

5.5. Ablations and efficiency

The development and probe runs provide directional evidence for the operator in Equation 3. A single token gave 25% new-answer preference on a 12-question probe, whereas a 4–16-token window reached 75%. On the 30-case CF split, one layer (22) gave 46.67% preference and 76.67% mismatched locality; three layers (22,24,26) gave 66.67% preference but 63.33% locality. Continuing the injection for two decoding steps reduced confirmation question accuracy from 38.67% to 34.67%; a full token trajectory reached only 17.0% on a development validation. These are ablations, not additional independent confirmation claims.

For the Qwen 80-case efficiency audit, vector construction averaged 2.273 s/case, base generation 0.388 s/pair, and steered generation 0.924 s/pair. No-router exact locality was 3.33%; the router costs 3.584 s/case in the 10,000-pair audit.

6. DISCUSSION AND LIMITATIONS

The experiments support three restrained conclusions. First, conditional activation intervention can improve multi-hop factual adaptation on two instruction-tuned models without changing their parameters. Second, routing produces high system-level locality, but the intervention itself is not local after a false route. Third, retrieval is a strong accuracy control and should be included in any comparison; Value-CQAG is preferable when selective, lightweight, and reversible adaptation is more important than maximizing answer accuracy.

The study has limitations. The confirmation pool has 100 cases and is not a broad pretraining-scale benchmark. The router uses lexical overlap and can miss paraphrases. The five-question parameter-editing locality protocol and 10,000-pair routed audit measure related but different quantities. Finally, the MEMIT implementation uses a hardware compatibility path (fp16 loading and CPU double solve) on a 24-GB GPU. We therefore report the exact protocol and do not claim an unmodified official MEMIT run. Remaining experiments: before a definitive submission claim, we need relation-type macro accuracy, independently sampled seeds and a second held-out pool, semantic/cross-lingual router tests, matched resource comparisons, and a trained locality-regularized intervention module. The reserved 60-case pool remains sealed and is intentionally not included.

7. CONCLUSION

We presented a cross-model study of locality-aware activation intervention for multi-hop factual adaptation. The key result is not a new accuracy record: it is a measured separation between adaptation quality, routing coverage, and unintended change. This separation makes failure analysis actionable and gives future work a concrete target: improve semantic routing and multi-hop composition while preserving the high locality of the gated path.

Acknowledgment

This work received no external funding. The authors thank the open-source model and benchmark communities for releasing the software and data used in this study.

Compliance with Ethical Standards

This study uses publicly available benchmark cases and pretrained models; it involves no human subjects, user data, or private or personally identifiable information. The evaluation is automated and reports aggregate outcomes. The reserved 60-case pool remains sealed and was not read or evaluated during this study.

8. REFERENCES

[1] Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy, "Transformer feed-forward layers are key-value memories," in Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, 2021, pp. 5484–5495.

[2] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov, "Locating and editing factual associations in GPT," in Advances in Neural Information Processing Systems, 2022.

[3] Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau, "Mass-editing memory in a transformer," in International Conference on Learning Representations, 2023.

[4] Eric Mitchell, Charles Lin, Antoine Bosselut, Chelsea Finn, and Christopher D. Manning, "Memory-based model editing at scale," in Proceedings of the 39th International Conference on Machine Learning, 2022, pp. 15817–15831.

[5] Zexuan Zhong, Yifan Wu, Michael D. Morse, Yiran Zhang, Steven H. Wang, et al., "MQuAKE: Assessing knowledge editing in language models via multi-hop questions," in Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, 2023.

[6] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen tau Yih, Tim Rocktaeschel, Sebastian Riedel, and Douwe Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Advances in Neural Information Processing Systems, 2020.

[7] Ce Zheng, Lei Li, Qingxiu Dong, Yuxuan Fan, Zhiyong Wu, Jingjing Xu, and Baobao Chang, "Can we edit factual knowledge by in-context learning?," in Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, 2023, pp. 4862–4876.

[8] Alexander Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Mark Goldstein, "Activation addition: Steering language models without optimization," arXiv preprint arXiv:2308.10248, 2023.

[9] Kang Li, Yutong Zhang, Ziqian Zhong, and Liwei Wang, "Inference-time intervention: Eliciting truthful answers from a language model," in Advances in Neural Information Processing Systems, 2023.